

TCMC • TAIWAN CHORAL MUSIC CENTER

Chord Voicing Trainer

User Manual

(和弦聲部音準練習)

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1. Overview

"Chord Voicing Trainer" is a web-based tool from the Taiwan Choral Music Center (TCMC) that lets you type any chord symbol, freely assign each chord tone to any of 8 voice parts (S1, S2, A1, A2, T1, T2, B1, B2) at any octave, pick one part as your singing target, and play the rest as accompaniment — with real-time microphone pitch detection checking whether you're matching the target.

Unlike the companion apps "Harmonic Third Pitch Trainer" and "Harmonic Seventh Pitch Trainer," this tool uses standard 12-tone equal temperament rather than Just Intonation. It's built for practicing arbitrary chord voicings and pitch recognition, not for training the blended, beat-free sound of Just-Intonation intervals.

Key Features

- Type any chord symbol (e.g. Am7(-5), G7, Cmaj7, C7(b9,#11)) and have all its chord tones parsed automatically
- Freely assign any chord tone at any octave (within A2–A5) to any of the 8 voice parts (S1/S2/A1/A2/T1/T2/B1/B2)
- Each voice part has its own play/mute toggle, so you can preview the voicing while still configuring it
- Pick one part as the singing target; freely toggle the rest on or off as accompaniment
- Real-time microphone pitch detection with a ±6 cent tolerance band, shown live on the chart
- Chinese / English interface toggle

Tip: headphones are recommended. Over speakers, the accompaniment can bleed into the mic and throw off the pitch judgement.

2. Chord Input

Type a chord symbol into the "Chord Input" field and press "Parse" (or just hit Enter). The app determines the root and chord quality, lists every chord tone, and labels each one with a numbered-notation scale degree (always relative to C major).

Below the parsed result, each chord tone appears as a clickable button (e.g. "🔊 6 • A") — tap it to hear that pitch immediately, which is handy for getting familiar with each tone before assigning voice parts.

Common Chord Symbols

Input	Chord Quality	Parsed Tones
C	Major triad	C, E, G
Cm	Minor triad	C, Eb, G
C7	Dominant 7th	C, E, G, Bb
Cmaj7	Major 7th	C, E, G, B
Cm7	Minor 7th	C, Eb, G, Bb
Cdim7	Diminished 7th	C, Eb, Gb, A
Cm7 (-5)	Half-diminished (m7b5)	C, Eb, Gb, Bb
C9	Dominant 9th	C, E, G, Bb, D
Csus4	Suspended 4th	C, F, G

Alteration Modifiers

Add parentheses after a chord symbol to alter or add extension tones — flat 5th, sharp 11th, flat 13th, and so on. The rule is simple:

- If the chord already has a tone at that scale-degree slot (e.g. the 5th), the modifier replaces it
- If the chord has no tone at that slot (e.g. an 11th or 13th), the modifier adds a new extension tone
- Multiple modifiers can be combined, comma-separated, e.g. (b9,#11)
- + and # both mean sharp and behave identically; - and b both mean flat and behave identically

Input	What happens	Parsed Tones
Cm7 (-5)	Replaces the existing 5th with a flat 5th	C, Eb, Gb, Bb
Cm7 (#11)	No 11th in the base chord, so a sharp 11th is added	C, Eb, G, Bb, F#
Cm7 (-13)	No 13th in the base chord, so a flat 13th is added	C, Eb, G, Bb, Ab
C7 (b9, #11)	Two alterations applied at once, comma-separated	C, E, G, Bb, Db, F#

Parsing a new chord resets any voice-part assignments you've already made — you'll need to reassign them.

3. Voice Part Assignment

Once a chord parses successfully, 8 dropdown menus appear — one for each voice part: S1, S2, A1, A2, T1, T2, B1, B2 (Soprano 1/2, Alto 1/2, Tenor 1/2, Bass 1/2). Each dropdown can be set to "Not used," or to a specific chord tone at a specific octave (e.g. "G3," "C4," "Eb4").

The available note range spans A2 to A5 (three octaves, covering both treble and bass ranges) — which octave each part sings is entirely up to you; no part is locked to a particular register.

You don't need to fill all 8 parts — assign only as many as you need. Two parts can also share the same pitch class an octave apart (e.g. S1 sings G4 while T2 sings G3, forming an octave doubling).

The first voice part you assign automatically becomes the singing target; you can change this in the next section.

4. Singing Target & Playback Controls

Once parts are assigned, an "Assigned Parts & Singing Target" list appears, showing each part's label, note name, scale degree, and frequency.

Setting the Target

Each row has a "Set as target" option — selecting it makes that part your singing target (highlighted in gold), and every other row then shows its own independent play/mute button.

Playback Buttons

-  Play Chord: plays every assigned part together (including the target) — good for hearing the full configured harmony at once
-  Play Target: plays only the target part, so you can re-check the pitch as many times as you like; press again to stop
- The Play/Mute button next to each non-target part independently toggles that part's accompaniment playback

-  Start: begins the practice sequence — briefly demos the target note, then plays whichever accompaniment parts you've enabled, and starts microphone detection
-  Stop: ends the session, stopping all playback and mic detection
-  Enable/Disable Mic: manually control whether your voice is being detected

All of the playback controls above already work during the configuration stage — you don't need to press "Start" first to preview sounds.

5. How to Use

1. Type the chord you want to practice into "Chord Input" and press "Parse."
2. Check the parsed chord tones — tap the degree buttons to preview each one.
3. In "Voice Part Assignment," pick a pitch and octave for each part you need.
4. In the "Assigned Parts" list, choose one part as your singing target.
5. Toggle other parts' playback as needed, or press " Play Chord" to hear the full harmony once.
6. Press " Start" — the app demos the target note, then plays your enabled accompaniment parts and turns on the mic.
7. Sing the target part, aligning with the gold target line on the chart (± 6 cent tolerance).
8. When you're done, press " Stop," or just change the chord/voicing to start the next round.

The first time you use the app, your browser will prompt for microphone permission — choose "Allow," or real-time pitch detection won't work.

6. Pitch Chart & Tuning Judgement

The chart's vertical range adjusts dynamically to fit whichever parts are currently assigned. Thick lines mark each assigned part's pitch; gold marks the current singing target.

Your live pitch trace appears in green when within ± 6 cents of the target, and red when outside that band — so you can judge intonation by color alone. Below the chart you'll also see the detected frequency, the offset in cents from the target, a text judgement (in tune / sharp / flat), and signal diagnostics (RMS, clarity) for more advanced reference.

The tolerance is fixed at ± 6 cents (about 6% of an equal-tempered semitone) — generally the range within which the ear can clearly notice a pitch discrepancy.

7. Troubleshooting

Microphone not responding / can't detect pitch

- Confirm your browser has allowed this page to use the microphone (check the permission icon in the address bar)
- If you're using an in-chat preview window, security restrictions may block the permission request — download the HTML file and open it directly in a browser instead
- Confirm the URL starts with https — a plain local file (file://) often can't get microphone access on some mobile browsers

Play Chord / Play Target has no sound

- Check that your device isn't muted and the browser tab hasn't been silenced by the system
- Some mobile browsers need one manual tap before they'll unlock audio playback at all — if there's no sound the first time, try tapping again

Pitch line jumps around or the judgement feels unstable while singing

- Headphones are recommended over speakers, to avoid the accompaniment bleeding into the mic and throwing off detection
- Sing in a quiet environment and keep a consistent distance from the microphone

Chord symbol won't parse

- Confirm it starts with A–G (optionally followed by # or b), e.g. "Bb7," "F#m7"
- Supported chord types cover common triads, 6th, 7th, 9th, and suspended chords, plus bracketed alteration modifiers; unusual or highly specialized jazz chord symbols may not be recognized — the app will show a clear error message in that case

8. Version History

Current version: v1.0

Production Team

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Nature of software: a web-based tool for choral voice-part rehearsal and self-practice